

Section 3: Visualizing information with RViz and launch files

AA274A - Principles of Robot Autonomy I

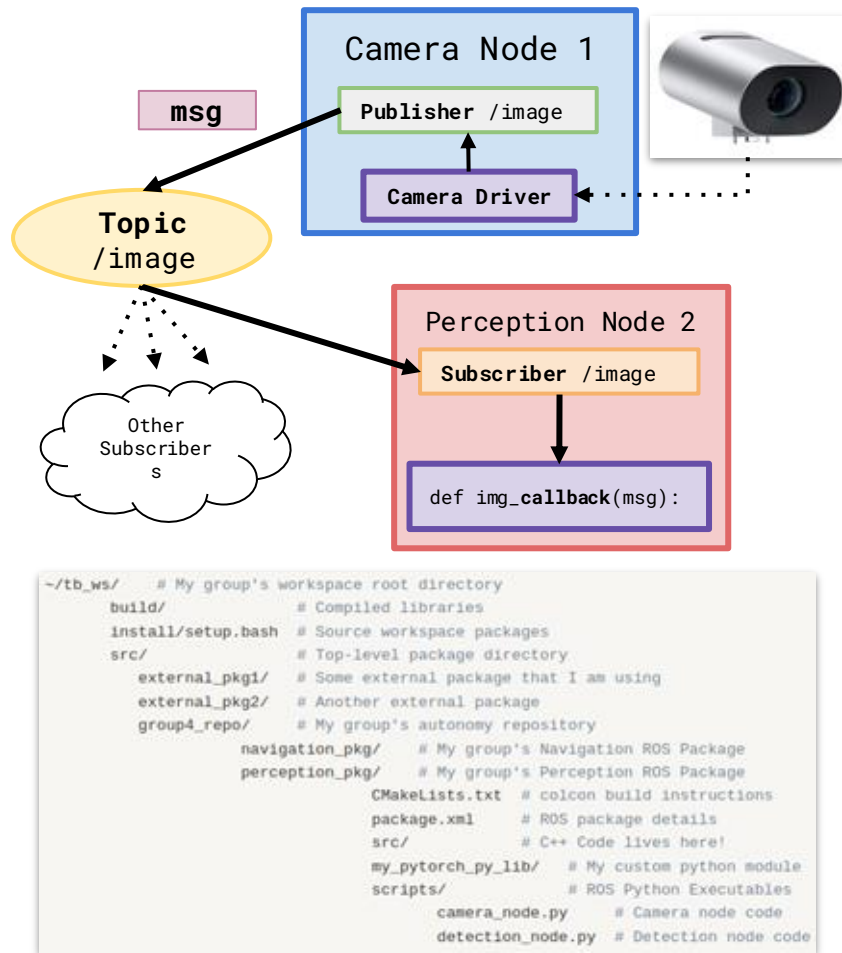
Week 3 – Autumn 2026

Section Attendance



Recap

- ROS development environment: workspaces, packages, CMakeLists.txt, package.xml
- ROS programming basics: nodes, topics and messages, publishers and subscribers.



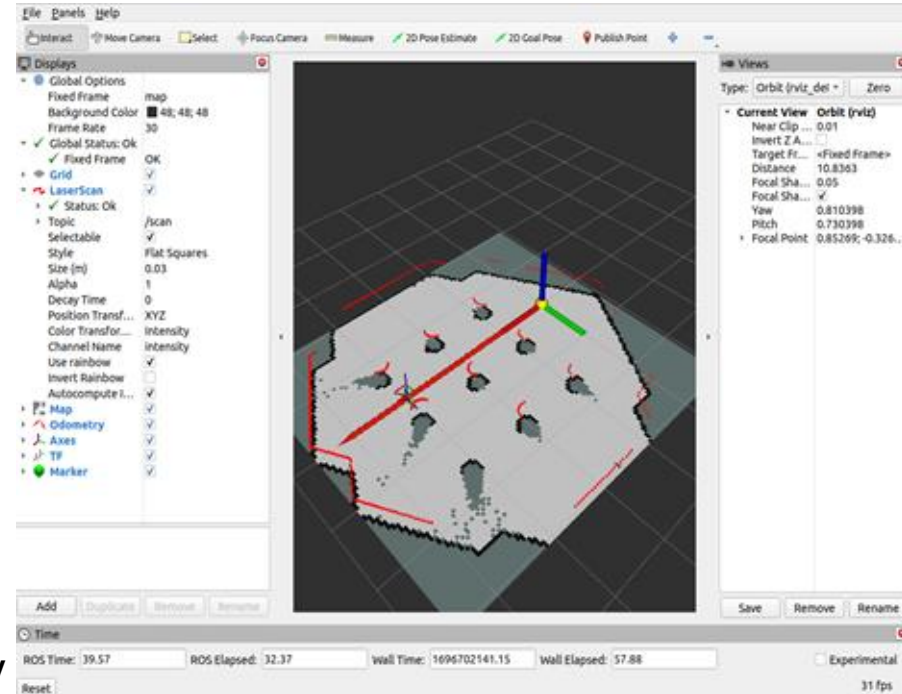
Today

- Become familiar with information visualization in ROS2 with RViz.
- Learn how to write launch files.
- Test out code on the actual robot!



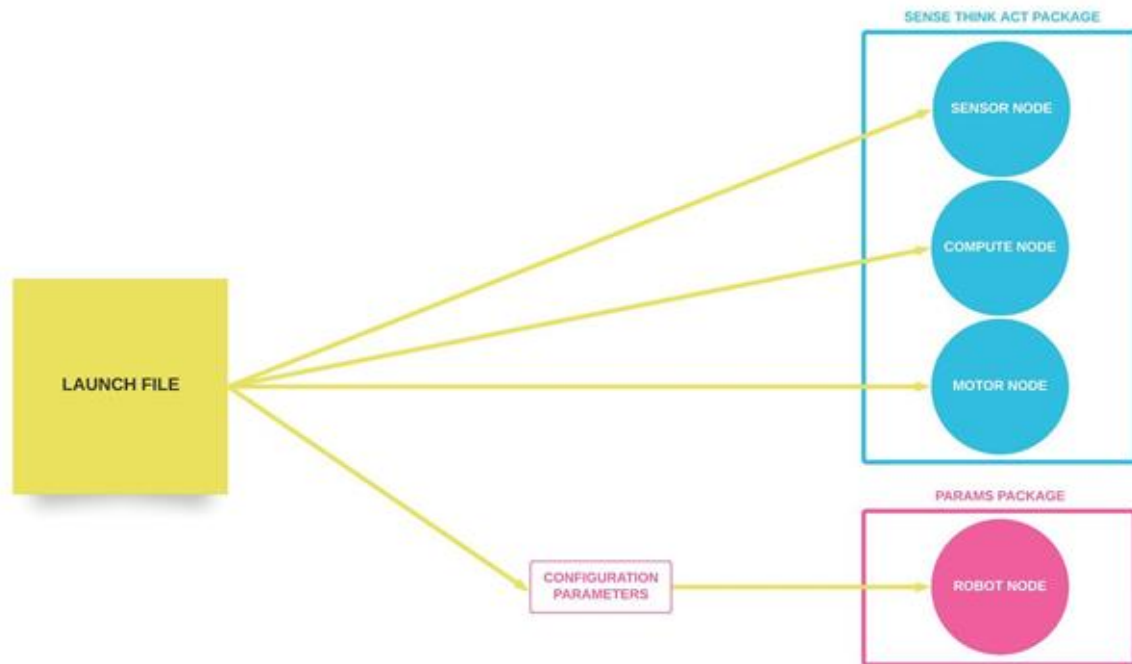
RViz

- ROS 3D visualization tool
- Can think of it as a graphical user interface (GUI) wrapper around `ros2 topic echo`.
- Visualizes information which otherwise wouldn't even be parsable, let alone parsable in context.
- E.g: Laser scans are a complicated mix of floating-point numbers, but RViz nicely plots them as point clouds which respect to the world frame.



Launch files

- Launch files simplify running complex systems with many nodes and specific configuration details.
- You can create launch files using Python, XML, or YAML, and run them using the `ros2 launch` command.
- We can also have launch files inside another launch file!



Live ROS Demo.

RViz Markers

- Say you have some intermediate goals or other world points that you use in your autonomy stack.
- Markers allow you to visualize these points aside from just printing them in the terminal.

